



## Drone Based Measurement And Drones' Certification

The deployment of drones/unmanned aerial vehicles (UAVs) has made the job easier, faster, and safer for surveyors. Some drones are specifically designed for data acquisition and surveying professionals, and are equipped with a high-resolution digital camera

Deepshikha Shukla

**I**n the era of the Internet of Things (IoT), one of the prevalent concerns is the deployment of unmanned aerial vehicles (UAVs) to automate different commercial applications, ranging from simple deliveries to more sophisticated aerial taxis. This creates a requirement for different measurements conducted with drones, which serve as the basis for further evaluation of potential UAV applications.

Surveying with drones makes the job easier, faster, and safer for surveyors. It captures and measures image data with a substantially increased spatial and temporal resolution as compared to many classical methods. Some drones are specifically designed for data acquisition and surveying professionals, who need accuracy and efficiency. These drones are equipped with a high-resolution digital camera that has been geometrically calibrated to perform measurements.

Drone mapping is the process of sur-

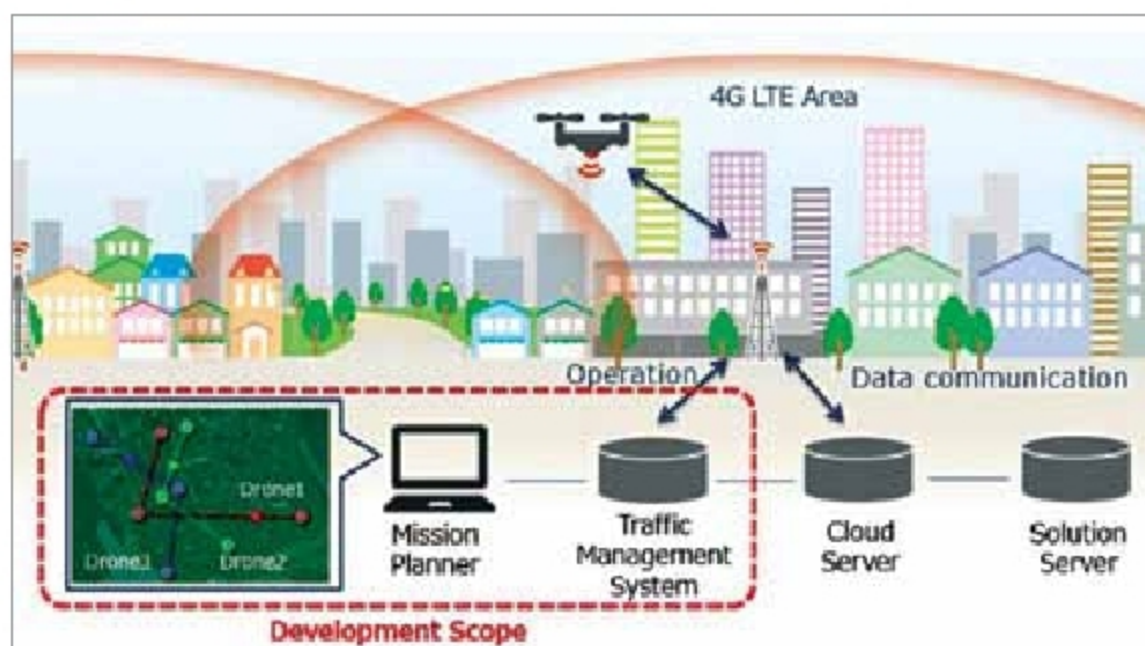
veying an area of land with a UAV, taking hundreds of pictures as it moves. Then with the help of computer software, it stitches and layers the images to analyse different results. Some of the most significant applications of drone mapping include construction sites, oil and gas, infrastructure, archaeological sites, mining, land surveying, property surveying, and landfill surveying.

### Communication network testing

To ensure safety, unmanned traffic management (UTM) systems are being developed to connect and operate multiple drones simultaneously. It is imperative to ensure that the quality of the UTM's communication network is reliable to have seamless control of the UAVs in the designated areas. This makes it critical to test the network quality at high altitudes before any UTM is successfully deployed or UAV flies in that area.

Testing the quality of service (QoS) and quality of experience (QoE) of LTE networks at flight altitude can help to identify the problem in the UAVs' path and altitudes such as a lack of data throughput, insufficient network coverage, and high latency, which can pose a problem in UTM's communication with the UAVs.

RF scanners and QualiPoc Android handsets mounted on a UAV can be used for testing network QoS and QoE. The coverage of a UAV's flight path at different altitudes can be determined by measuring and plotting network coverage parameters such as reference signal received power (RSRP) and signal to interference plus noise ratio.



4G LTE control system for drones (Credit: [www.terra-drone.net](http://www.terra-drone.net))